



Team Details

- **Team:** Superiores
- **Members:**

Christopher Pérez Cortés: A recent addition to the Robotics Club at the "Profr. Moisés Sáenz Garza" Escuela Normal Superior; he joined the team after completing an intensive robotics course. Currently in his third year of secondary school, he possesses skills in electronics, programming, and 3D modeling; this is his first time participating in a WRO tournament.

Bárbara Daiana García Balboa: A member of the Robotics Club at the "Profr. Moisés Sáenz Garza" Escuela Normal Superior for the past six months; she joined the team to participate in WRO 2026 for the first time. She became a member after completing two intensive robotics training courses.

- **Programming Language:** C++
- **Microcontroller:** Arduino Uno
- **Sensors:** 5 ultrasonic (2 right, 2 left, 1 front), open camera (OpenMV)
- **3D Design:** Tinkercad

Entry 1 — Early March

Stage: Preparation for online tournament

The team began its preparation aiming for an online tournament. In this first stage, the robot was built entirely with LEGO pieces.

In the online tournament, the team managed to complete the Open Challenge, but failed to complete the Obstacle Challenge (the challenge that involved dodging green and red blocks).

As the weeks went by, it was detected that the LEGO structure took up too much space, which made the internal organization of the components (motors, wiring, sensors) difficult. As a solution, the team decided to migrate toward 3D design and printing of parts (modeled in Tinkercad), achieving a more compact and better-organized structure than the original LEGO version.

Entry 2 — Weekly progress (hardware improvements)

Stage: Iterative robot improvements

As the weeks progressed, several key improvements were made to the robot's components:

- **Camera:** a HuskyLens was initially used; it was replaced by an open camera (OpenMV), seeking better performance for obstacle/color detection.
- **Servomotor:** a 2.2 kgf Steren servomotor was swapped for an SG90, adjusting the steering system to the new design needs.

Incidents and solutions:

- The 3D-designed parts did not fit correctly with the motors and the servomotor. Solution: the parts were sanded down to give them the necessary clearance.
- To implement the Ackermann steering strategy, a LEGO piece was included as part of the mechanism; additionally, two pieces were cut and an extra 3D-printed piece was manufactured to complete the assembly.
- The circuit board (PCB) burned out on one occasion because the batteries were inserted backward. Solution: a new board was installed and completely re-soldered.

Detection and control components:

- **5 ultrasonic sensors:** two on the right, two on the left, and one in front.
- **Microcontroller:** Arduino Uno.

Entry 3 — Week before the competition (current)

Stage: Final testing, documentation, and backups

One week away from the national competition in Tijuana, the team is in the home stretch:

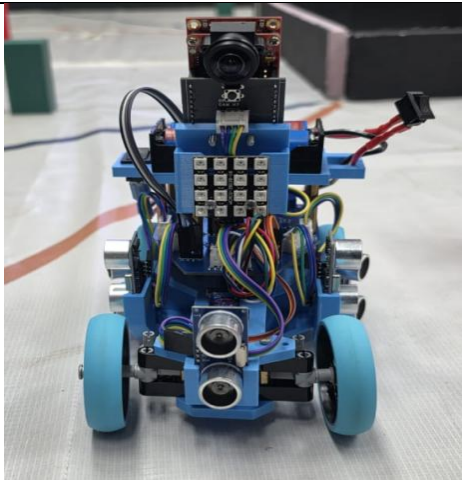
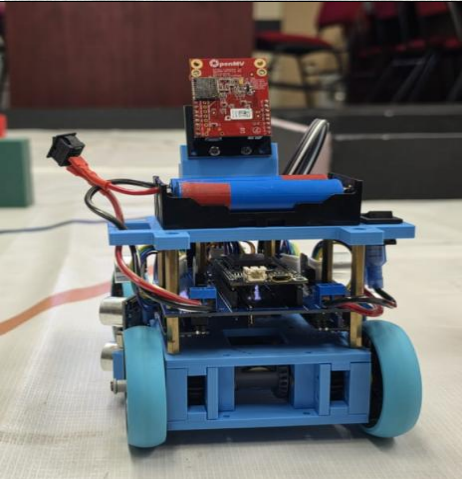
- **Programming (C++):** the programming logic is already working correctly, both for the Open Challenge and the Obstacle Challenge. Pending: fine calibration so the robot returns exactly to its starting position (it currently drifts a bit).
- **Documentation:** technical documentation for the project is being drafted and organized.
- **Spare parts:** spare cables, 3D-printed parts, and other components are being manufactured as a contingency plan in case of failures during the competition.

- **Support:** the team has the backing of an advisor professor throughout the entire process.

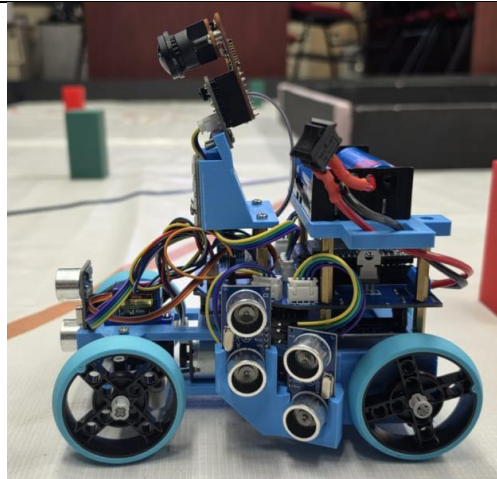
Plan for the upcoming days:

- Continue programming tests to verify that the entire system works consistently.
 - Keep manufacturing spare parts for critical components.
 - Fine-tune the robot's calibration to eliminate trajectory deviations.
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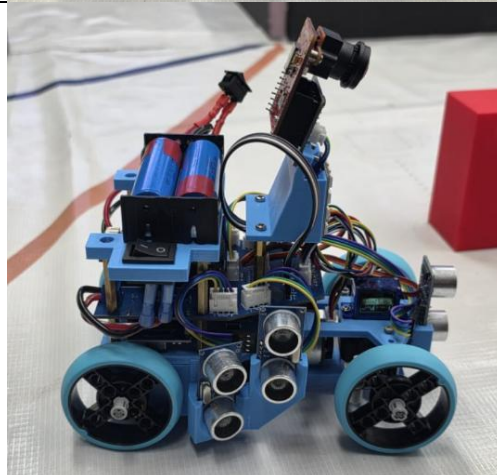
Robot Photos :

Front	
Back	

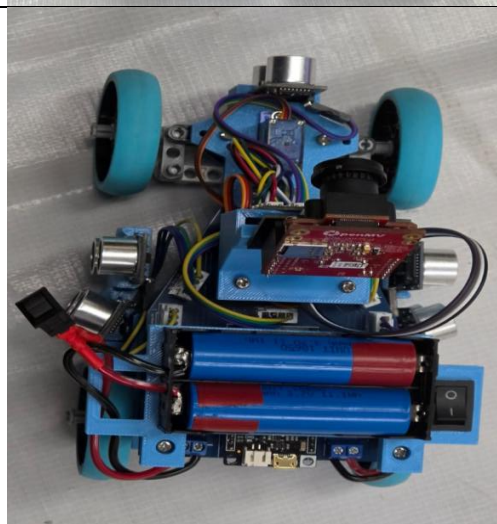
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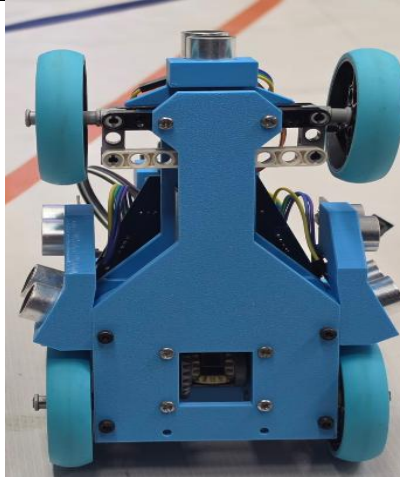
Right



Top



Bottom



We are currently striving to become an ever-larger team, and we eagerly await the day of the competition.

- Superiores